

Fatemeh Chitsaz

✉ f.chitsaz7918@gmail.com ☎ +98-9373447523 🔗 Google Scholar in LinkedIn 🐙 GitHub

Summary

Data Scientist with a strong background in software development and machine learning, specializing in NLP, anomaly detection, and time-series analysis. Experienced in tackling real-world challenges, from financial fraud detection to stock market prediction, using advanced statistical and AI-driven approaches. Passionate about building efficient, data-driven solutions and optimizing machine learning models for impactful results.

Education

University of Tehran

2023 – present

MS in Information Technology

- GPA: 3.8/4.0
- **Coursework:** Machine Learning, Deep Learning, Natural language Processing, Speech Processing, Image Processing

Alzahra University

2019 – 2023

BS in Computer Engineering

- GPA: 3.8/4.0
- **Coursework:** Programming, Data Structures and Algorithms, Databases, Image Processing, Distributed Systems, Big Data, Cloud Computing

Research

- **Dual Adaptation of Time-Series Foundation Models** – Proposed a dual-module adaptation framework (Generalizer Adapter + Identity Signature) for large-scale time-series foundation models, enhancing stability, cross-domain generalization, and zero-shot forecasting performance in noisy, low-resource financial environments.
- **Sequential Layer-Wise Fine-Tuning** – Developed a contribution-based sequential fine-tuning methodology that updates model components in an empirically determined order of importance, mitigating overfitting and improving transferability to previously unseen datasets.
- **Hybrid Multivariate–Univariate Forecasting (StockFM)** – Designed a hybrid architecture integrating pre-trained univariate foundation models with multivariate predictors, achieving substantial gains in predictive accuracy (MSE reduction) and directional forecasting (macro-F1 improvement) for equity markets.
- **LLM–Time Series Integration** – Investigated synergies between large language models and time-series forecasting systems, including the adaptation of architectures such as Time-LLM and Tempo to capture complex temporal–semantic dependencies.
- **RL-Enhanced Financial Forecasting** – Fine-tuned large language models on conversational financial data, incorporating self-reflection mechanisms and Proximal Policy Optimization (PPO) to enhance policy learning for stock movement prediction.
- **Parameter-Efficient Model Adaptation** – Applied and extended parameter-efficient tuning strategies (e.g., LoRA variants, bias-only tuning, layer-norm adaptation) to create lightweight, computationally efficient adaptation pipelines for high-noise domains.
- **Foundation Model Adaptation under Data Scarcity** – Developed architectures and training strategies enabling robust adaptation of time-series foundation models with limited supervision and multi-variable data, improving resilience and accuracy in volatile market conditions.

Publications

Dual Adaptation of Time-Series Foundation Models for Financial Forecasting

May 2025

Fatemeh Chitsaz, Saman Haratizadeh [ICML 2025](#) [🔗](#)

Learning What Matters First: Sequential Adaptation of Time Series Foundation Models for Robust Financial Forecasting

May 2025

Fatemeh Chitsaz, Saman Haratizadeh [ICML 2025](#) [🔗](#)

Bridging Univariate Foundation Models and Multivariate Financial Signals for Stock Price Prediction

March 2025

Fatemeh Chitsaz, Saman Haratizadeh [Journal of security and computing](#) [🔗](#)

Work Experience

Senior Data Scientist

Muscat, Oman (remote)

Unity Global Industries

Feb 2025 – Present

- Developed a deep learning-based platform for forecasting stock market behavior using time series models, providing evaluation metrics and trading performance benchmarks for comparing different predictors
- Integrated Explainable AI (XAI) techniques to interpret model predictions and provide transparent insights into market status, enhancing trust and decision-making support

Senior Data Scientist

Tehran, Iran

BehSazan Mellat (Subsidiary of Mellat Bank)

Sep 2024 – Feb 2025

- Worked on unsupervised learning methods and autoencoders for fraud and money laundering detection
- Utilized ADMoE (Adaptive Mixture of Experts) from Microsoft for noisy label

Senior Software Engineer

Tehran, Iran

Dotin (Subsidiary of Passargad Bank)

Jul 2023 – Sep 2024

- Developed an ERP system for Pasargad Bank
- Designed and implemented a payroll slip application for Sepah Bank

Software Engineer

Tehran, Iran

Institute of Applied Intelligent Systems, (Subsidiary of University of Tehran)

Jul 2021 – Jul 2023

- Developed a messenger application using Flutter, Java, and Scala
- Designed and implemented video/audio call solutions using WebRTC.
- Developed a bot platform (Bot Father) for automated chatbot development

Teaching Assistant

Big Data

Tehran, Iran

University of Tehran, [Dr. Saman Haratizadeh](#) [🔗](#)

Feb 2025 – Aug 2025

- Supported coursework in Big Data, helping students build scalable data pipelines using Apache Spark, Hadoop, and MapReduce.

Speech Processing

Tehran, Iran

University of Tehran, [Dr. Hadi Veisi](#) [🔗](#)

Feb 2025 – Aug 2025

- Mentored students on Speech Processing projects involving feature extraction, acoustic modeling, and neural architectures for voice-based applications

Machine Learning

Tehran, Iran

University of Tehran, [Dr. Saman Haratizadeh](#) [🔗](#)

Sep 2024 – Feb 2025

- Assisted in teaching Machine Learning concepts including supervised learning, unsupervised clustering, and deep learning frameworks
- Guided students in hands-on implementation using Python, PyTorch, and TensorFlow.

Technical Competencies

Programming Languages: Python, Java, Dart, C

Frameworks & Libraries: PyTorch, TensorFlow, JAX, HuggingFace Transformers, LangChain, Flutter

Data Management Systems: SQL, PostgreSQL, Apache Cassandra

Development & Deployment Tools: Git, Docker, gRPC, WebRTC, Google Firebase, REST API